

Ralph Balazs

Software Engineer Intern • 6-month internship, available from September 2026 • Paris, France
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TECHNICAL SKILLS

- **Programming:** Python, C/C++, TypeScript, PHP, SQL, Bash
- **Backend, APIs & Frontend:** FastAPI, OAuth 2.0 / JWT, REST APIs, React Native
- **Computer Vision & Robotics:** YOLO (Ultralytics), OpenCV, LeRobot (Hugging Face), NVIDIA Jetson
- **Databases:** PostgreSQL, MySQL
- **DevOps & Deployment:** Docker, Kubernetes (K3s, K3d), GitOps (Argo CD), CI/CD, Linux, Git
- **Testing & Tools:** gdb, valgrind, Make
- **Spoken languages:** French (native) • English (fluent) • Spanish (basic)

EDUCATION

École 42 – Software Engineering	Paris, France
<i>RNCP Level 7 (Master's equivalent) in progress, AI & Algorithms</i>	November 2023 – Present
Lycée de l'Europe	Dunkerque, France
<i>CPGE PTSI – Physics, Technology & Engineering Sciences</i>	2017 – 2018

PROJECTS

Autonomous Chess Robot 42AI, Paris – team project (in progress)

Python, LeRobot, YOLO, OpenCV, NVIDIA Jetson

- Built a computer-vision module in Python (YOLO/Ultralytics, OpenCV) for real-time object detection on a physical robot setup
- Calibrated a SO-101 arm and recorded demonstration datasets via LeRobot teleoperation for an imitation-learning pipeline
- Installed and configured an NVIDIA Jetson Orin NX as the onboard compute unit driving the robotic arm

Music Room – Collaborative Music Mobile App 42 School, Paris – team project

FastAPI, PostgreSQL, React Native (Expo Go), Docker

- Implemented Google OAuth 2.0 (Authorization Code) on FastAPI with server-side code exchange, issuing custom JWT access/refresh tokens with rotation
- Built the full auth module (email/password with bcrypt, OAuth linking, anti-CSRF state cookies) and integrated Google login on the React Native mobile client

Matcha – Private Messaging 42 School, Paris – team project

PHP, MySQL, JavaScript, Docker

- Implemented private messaging in PHP/MySQL for matched users only, with server-side match and block verification, CSRF tokens, and prepared statements (PDO)
- Built near-real-time message and notification sync in vanilla JavaScript via incremental polling to limit server load

Kubernetes GitOps Pipeline 42 School, Paris – team project

K3s / K3d, Argo CD, Docker, Vagrant, Bash

- Designed and deployed a Kubernetes infrastructure (K3s, Traefik) hosting 3 internal web services, with Ingress-based routing
- Built a GitOps pipeline with Argo CD to automatically sync manifests from GitHub

EXPERIENCE

École 42	Paris, France
<i>Tutor</i>	March 2025 – September 2025

- Graded 48-hour team projects (Rush) and proctored exams during 42 piscines applying strict evaluation criteria
- Provided feedback on code quality, edge cases, and security; helped students debug segfaults, race conditions, and memory leaks

Carré Clay Association & École 42 Chess Club	Paris, France
<i>Chess Instructor & Club President</i>	October 2025 – June 2026

- Teach chess classes to groups of around 30 students, lead corporate team-building events (Deloitte), and run the École 42 chess club

SGS	Saint-Nazaire, France
<i>Marine Surveyor</i>	September 2020 – July 2023

- Conducted cargo and container inspections and sample analysis; managed inspection data and reporting